

# Errata: *Generalizing geometric nonwindowed scattering transforms on compact Riemannian manifolds*

Everything below is keyed to the published version, *Sampling Theory, Signal Processing, and Data Analysis* **22**, 19 (2024), <https://doi.org/10.1007/s43670-024-00093-x>, and numbering is that of the published article. The earlier preprint arXiv:2402.10771 is an initial draft and numbers results differently.

Items **A1–A5** change what is stated; **B** lists proof corrections that leave statements intact; **C** lists typos and is set in smaller type. None of the corrections changes the range  $1 < q \leq 2$  or the conclusions of the paper.

This errata was prepared by Claude Opus 5.0. The authors have checked the corrections and take responsibility for all results.

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## A. Statement changes

### A1. Theorem 1 holds modulo the constant component

Theorem 1 states  $\sum_{j \in \mathbb{Z}} \|f * \psi_j\|_2^2 = C^2 \|f\|_2^2$ . The telescoping in the proof gives  $\sum_j |\hat{\psi}_j(k)|^2 = C^2$  only when  $\lambda_k > 0$ . For  $k = 0$  we have  $\lambda_0 = 0$ , so

$$\hat{\psi}_j(0) = [|G(0)|^2 - |G(0)|^2]^{1/2} = 0 \quad \text{for every } j \in \mathbb{Z},$$

and the corresponding sum is 0 rather than  $C^2$ . Hence the correct statement is

$$\sum_{j \in \mathbb{Z}} \|f * \psi_j\|_2^2 = C^2 \left( \|f\|_2^2 - |\hat{f}(0)|^2 \right) \leq C^2 \|f\|_2^2,$$

with equality precisely when  $\hat{f}(0) = 0$ . Only the inequality is used later in the paper, so no subsequent result is affected.

The remark following Theorem 1 should be read accordingly: for  $G(\lambda) = e^{-\lambda}$  one has  $C = 1$  and the relation is an isometry on the functions with  $\hat{f}(0) = 0$ , not on all of  $\mathbf{L}^2(\mathcal{M})$ .

### A2. Corollary 10 is an inequality

Corollary 10 states  $\|\bar{S}_2^m f\|_{\ell^2(\mathbb{Z}^m)}^2 = C^{2m} \|f\|_{\mathbf{L}^2(\mathcal{M})}^2$ . This should read  $\leq$ . Beyond the defect in A1 at the first layer, equality fails strictly at every later layer: the input to layer  $k+1$  is  $U[j_1, \dots, j_k]f \geq 0$ , which has strictly positive integral whenever it does not vanish. Applying A1 one layer at a time gives, for instance,

$$\sum_{j_2 \in \mathbb{Z}} \|U[j_2]U[j_1]f\|_2^2 = C^2 \left( \|U[j_1]f\|_2^2 - \text{vol}(\mathcal{M})^{-1} \|U[j_1]f\|_1^2 \right),$$

where the subtracted term is the square of the corresponding  $q = 1$  scattering coefficient.

### A3. The regularity condition should be $c_2/c_1 < 2$

Lemma 4, and the statements that quote it (Theorem 6, Corollary 8, Theorem 12, Corollary 13, Theorem 14, Theorem 18), assume  $c_1 c_2 < 2$ . The condition that the proof requires is

$$\frac{c_2}{c_1} < 2.$$

In the proof of Lemma 4, item 2 of Lemma 3 gives  $|y - z| \leq c_1^{-1} r(y, z)$ , so

$$r(y, w_x) \leq c_2 |y - w_x| \leq c_2 |y - z| \leq \frac{c_2}{c_1} r(y, z),$$

and with  $r(y, z) \leq \frac{1}{2} r(x, y)$  the conclusion needs  $1 - \frac{c_2}{2c_1} > 0$ . The remaining occurrences of  $c_1 c_2$  in that proof should be replaced by  $c_2/c_1$  in the same way.

The distinction matters beyond bookkeeping. The product  $c_1 c_2$  is not invariant under rescaling the chart maps  $\varphi_i$ : replacing  $\varphi_i$  by  $a\varphi_i$  sends  $c_1 c_2$  to  $c_1 c_2 / a^2$ , so the stated condition could be met by reparametrizing. The ratio  $c_2/c_1$  is invariant, satisfies  $c_2/c_1 \geq 1$  always, and measures the distortion of the metric in the coordinates of the covering.

Example 1 is unaffected: there  $c_1 = 1$ , so  $c_2/c_1 = c_2$  and both computations still give a value below 2.

### A4. The Schwartz condition belongs to the wavelet generator

Theorem 6, Corollary 8, Theorem 12, Corollary 13 and Theorem 14 assume  $G \in \mathcal{S}(\mathbb{R}^+)$ , but the hypothesis that Lemma 2 actually requires is on the function generating the wavelets. Writing

$$F(u) := [|G(u/2)|^2 - |G(u)|^2]^{1/2}, \quad \hat{\psi}_j(n) = F(2^j \lambda_n),$$

Lemma 2 applies when  $F \in \mathcal{S}(\mathbb{R}^+)$ , and  $F(0) = 0$  holds automatically. The condition  $G \in \mathcal{S}(\mathbb{R}^+)$  does not imply it: the difference may vanish to first order at the origin, in which case  $F$  behaves like  $u^{1/2}$  and is not differentiable at  $u = 0$ . This happens for  $G(u) = e^{-u}$ , where  $F(u)^2 = e^{-u} - e^{-2u} = u + O(u^2)$ , which is exactly the choice named in the remark after Theorem 1.

A sufficient condition is that  $|G(u/2)|^2 - |G(u)|^2 = u^2 h(u)$  for some  $h \in \mathcal{S}(\mathbb{R}^+)$  that is positive on  $[0, \infty)$ , since then  $F(u) = u h(u)^{1/2}$ . The Gaussian  $G(u) = e^{-u^2}$  satisfies this:  $e^{-u^2/2} - e^{-2u^2}$  is even, vanishes to second order at the origin and is positive on  $(0, \infty)$ .

### A5. Missing hypothesis and constants in the stability theorems

Theorem 16 is stated for  $f \in \mathbf{L}^2(\mathcal{M})$ , but the bound contains  $\lambda^n$  and the proof invokes Lemma 15; the hypothesis that  $f$  be  $\lambda$ -bandlimited is needed and should be added to the statement.

In both Theorem 16 and Theorem 18 the constant depends on the number of layers. Theorem 9 contributes a factor  $C^m$  and Theorem 12 a factor  $C_q^{m/q}$ , so the constants should be written  $C_m(\mathcal{M})$  and  $C_{m,q}(\mathcal{M})$ . In the proof of Theorem 18, the displayed constant  $C_q$  should be  $C_q^{m/q}$ .

## B. Proof corrections (statements unchanged)

### B1. The inductions in Theorems 9 and 12

Both proofs reach a sum of *norms of differences* and then identify it with a scattering norm, which is a sum of *differences of norms*. In Theorem 9 the step reads

$$C^2 \sum_{(j_1, \dots, j_k) \in \mathbb{Z}^k} \|U[j_1, \dots, j_k]f - U[j_1, \dots, j_k]g\|_2^2 = C^2 \|\bar{S}_2^k f - \bar{S}_2^k g\|_{\ell^2(\mathbb{Z}^k)}^2,$$

and Theorem 12 has the analogous line with  $\|\cdot\|_q$ . These two quantities are not equal, and the reverse triangle inequality bounds the second by the first, so the induction hypothesis does not cover the quantity produced by the step.

**Correction.** Run the induction on the larger quantity. For  $m \geq 1$  set

$$E_m(f, g) := \sum_{p \in \mathbb{Z}^m} \|U[p]f - U[p]g\|_2^2, \quad E_m^{(q)}(f, g) := \left( \sum_{p \in \mathbb{Z}^m} \|U[p]f - U[p]g\|_q^2 \right)^{q/2},$$

so that the reverse triangle inequality gives  $\|\bar{S}_2^m f - \bar{S}_2^m g\|_{\ell^2(\mathbb{Z}^m)}^2 \leq E_m(f, g)$  and  $\|\bar{S}_q^m f - \bar{S}_q^m g\|_{\ell^2(\mathbb{Z}^m)}^q \leq E_m^{(q)}(f, g)$ . Since the modulus is nonexpansive,

$$\begin{aligned} E_{k+1}(f, g) &= \sum_{p \in \mathbb{Z}^k} \sum_{j_{k+1} \in \mathbb{Z}} \left| \|U[p]f * \psi_{j_{k+1}}\| - \|U[p]g * \psi_{j_{k+1}}\| \right|_2^2 \\ &\leq \sum_{p \in \mathbb{Z}^k} \sum_{j_{k+1} \in \mathbb{Z}} \|(U[p]f - U[p]g) * \psi_{j_{k+1}}\|_2^2 \\ &\leq C^2 E_k(f, g), \end{aligned}$$

by Theorem 1 applied to  $U[p]f - U[p]g$ , with  $E_0(f, g) = \|f - g\|_2^2$ . The same computation with Corollary 8 in place of Theorem 1 gives  $E_{k+1}^{(q)} \leq C_q E_k^{(q)}$ . The stated bounds follow. The chains already written in the two proofs are otherwise unchanged.

### B2. Smaller items in the proof of Theorem 6

- In the display bounding the case  $2r_i \geq \delta$ , the middle expression carries  $(1 + 2^j r(x, x_i))^{n+1}$ ; the exponent should be  $2^{j/2}$ , as in the lines before and after it.
- In the conclusion of the estimate,  $\|Wb_i\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^1(B(x_i, r_i)^c)}$  should be  $\|Wb_i\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^1(B(x_i, 2r_i)^c)}$ , matching the exceptional set used throughout.

### B3. Lemma 17

The displayed chain bounds  $|f(x) - V_\xi f(x)|$  pointwise, while the statement is an  $\mathbf{L}^q(\mathcal{M})$  bound; the passage from one to the other contributes a factor  $\mu(\mathcal{M})^{1/q}$ , which is absorbed into  $C_{q, \text{Vol}(\mathcal{M})}$  but is not shown.

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## C. Typos

### Section 2

- Notation: “the maximum placement of  $\gamma$ ” should be displacement; “We use the notation  $\text{Isom}(\mathcal{M}_1, \mathcal{M}_2)$  be the set” is missing a verb.
- §2.2: “a notation of translation does not necessarily exist, To motivate...” should be “a notion of translation does not necessarily exist. To motivate...”. In the same passage, the basis  $\{c_n(x)\}_{n \geq 0}$  should be  $\{e_n(x)\}_{n \geq 0}$ , “fourier” should be capitalized, and the display stops short of the point being made, namely  $\langle f * g, e_n \rangle = \hat{f}(n)\hat{g}(n)$ .
- “since the the geometry of  $\mathcal{M}$ ” has a repeated word.
- Equation (5):  $|G(2^{j-1}n)|^2 - |G(2^j n)|^2$  should read  $|G(2^{j-1}\lambda_n)|^2 - |G(2^j \lambda_n)|^2$ .
- Equation (12): the  $m$ -layer windowed propagator is written with the same symbol as the unwinded one; it should be  $U_J[j_1, \dots, j_m]f := U_J[j_m] \cdots U_J[j_1]f$ .
- The display before (13) is missing  $f$  on the left:  $S_J[j_1, \dots, j_m]f = A_J U_J[j_1, \dots, j_m]f$ . In (13) itself,  $S_J$  should be  $S_J$ .
- Equation (14):  $S[j_1, \dots, j_m]f$  should be  $S_J[j_1, \dots, j_m]f$ , and the right-hand side carries a factor  $G(0)$ , which is 1 only under the normalization  $C = 1$ .
- Contributions list: “aribtrary”.

### Sections 3–5

- Section 3 title and Theorem 11: “littlewood paley” should be “Littlewood–Paley”.
  - “Caulderon-Zygmund” in the proof of Theorem 6, and “Holder” in the proof of Lemma 17.
  - After (13): “stable to diffeomporhisms”.
  - Theorem 9, proof: the delimiters in  $||f * \psi_j||_2 - |g * \psi_j||_2|^2$  are mismatched;  $\psi_{j+1}$  should be  $\psi_{j_{k+1}}$  in two places;  $U[j_1, \dots, j_n]g$  should be  $U[j_1, \dots, j_k]g$ ; and the hypothesis is assumed for  $n$  but then invoked “for some  $k \geq 1$ ”.
  - Theorem 12, proof:  $\ell^2(\mathbb{Z}^m)$  should be  $\ell^2(\mathbb{Z})$  in the  $m = 1$  display and  $\ell^2(\mathbb{Z}^{m+1})$  in the  $m + 1$  display; the line beginning “=  $C_q$ ” should be an inequality; and in the final display  $\|\cdot\|_{\mathbf{L}_{\ell^2(\mathbb{Z})}^q(\mathcal{M})}$  should be  $\|\cdot\|_{\mathbf{L}^q(\mathcal{M})}$ .
  - Conclusion: “manifolds that are not restricted the conditions” is missing “to”.
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## D. Unaffected

The framework of Sections 2 and 3, the weak  $(1, 1)$  estimate of Theorem 6 and its interpolation in Corollary 8, the isometry invariance of Theorems 11 and 14, and the boundedness and stability results for  $q \in (1, 2]$  all stand as stated, with the hypotheses as corrected in A3 and A4.